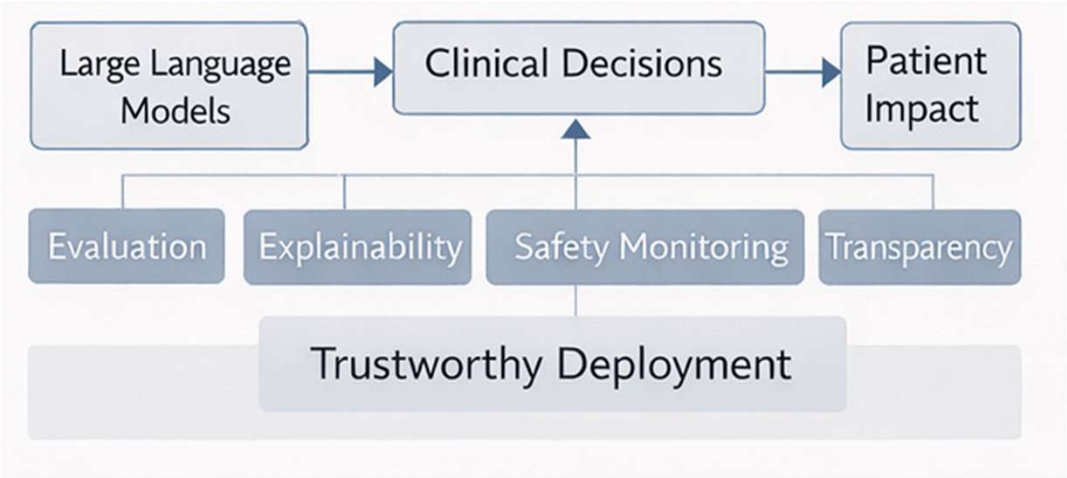
Trustworthy And Safe AI For Clinical Systems

Motivation

- Large language models increasingly support clinical summarization, diagnosis, and decisions.
- Current systems remain opaque, prone to hallucinated reasoning.
- Safety risks arise from unreliable outputs, inconsistent reasoning, and unverifiable decisions.
- Clinical settings require transparency, traceability, and explicit failure analysis.



Goal

Reliable, explainable, and traceable AI systems for responsible deployment

Trustworthy And Safe AI For Clinical Systems

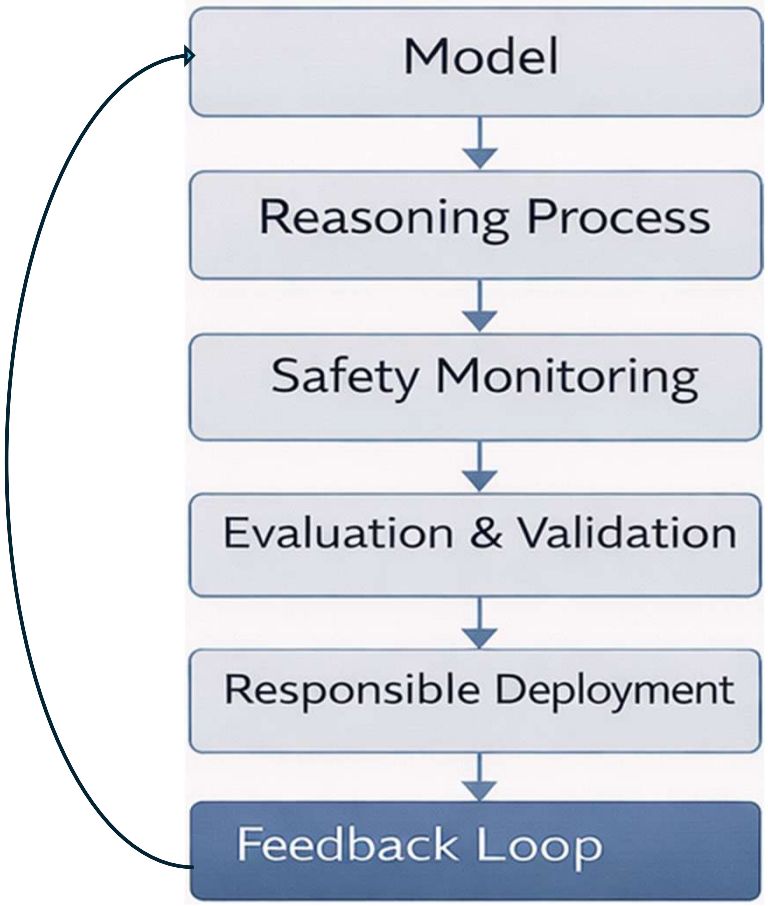
Trustworthy AI

- Reliability: stable behavior under input variation.
- Safety: prevention of harmful or unsafe recommendations.
- Explainability: interpretable reasoning and decision traceability.

Why Trustworthiness matters

- Clinical AI operates in high-risk environments with direct patient impact.
- Model outputs must be reliable, interpretable, and verifiable.
- Incorrect reasoning with correct predictions remains unsafe.
- Deployment requires safeguards against hallucinations and failures.

Trustworthy AI Architecture



Structured Fact Extraction for Reliable Clinical Table Summarization

Problem

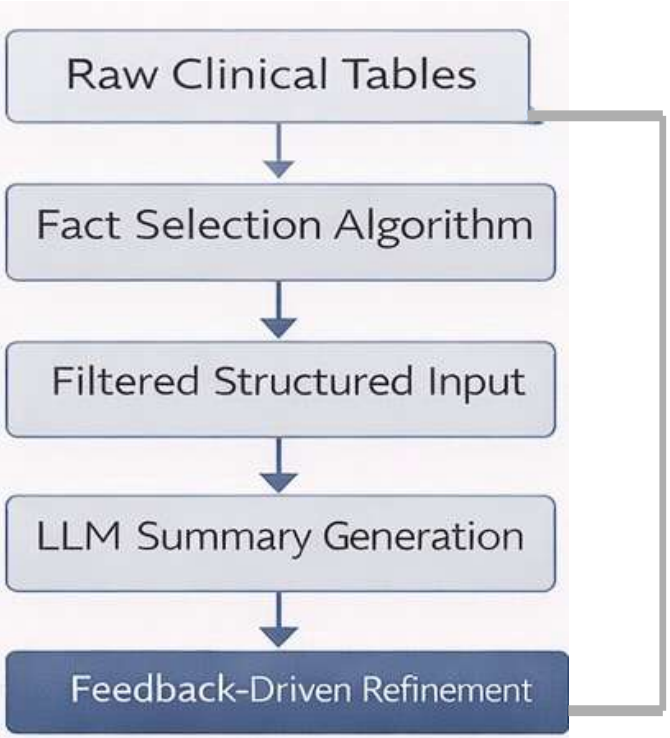
- Clinical study reports contain large structured tables with dense statistical and medical information.
- Direct prompting of LLMs produces incomplete or inconsistent summaries due to context length limits.
- Models frequently omit critical facts or generate imprecise clinical statements.

Key Idea

- Decompose table-to-text generation into fact selection and controlled generation.
- Extract clinically relevant information prior to summary generation.
- Reduce input complexity to improve factual consistency and precision.

Contribution

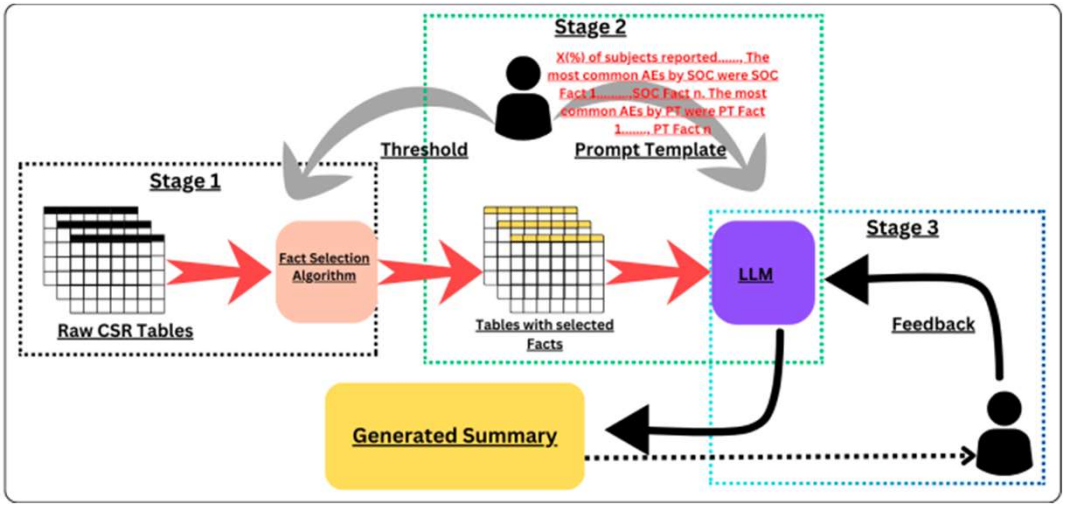
- End-to-end pipeline integrating fact selection and feedback loop
- Improves factual completeness and reduces hallucination content.
- Enables more reliable and clinically usable summaries.



Method: CSR Table Summarization Pipeline

Task

- Generate summaries from large clinical trial tables.
- Preserve and extract all important clinical facts.
- Avoid hallucinations.
- Integrate a feedback loop, allowing users to refine the generated summaries.



Pipeline Steps

- **Table Linearization:** convert structured tables into sequence format for LLM input.
- **Fact Selection:** extract important facts using algorithm.
- **Input Reduction:** remove irrelevant or low-frequency data.
- **Summary Generation:** LLM generates structured clinical summary.
- **Feedback Loop:** users refine output for accuracy and clarity.

Results: CSR Table Summarization Pipeline

Evaluation

- Used Claim Recall (completeness) and Claim Precision (conciseness).
- Used GPT-4o as LLM as a judge.
- Human evaluation using internal medical writer.
- Checked rule compliance, accuracy, conciseness, fluency.

Quantitative Results

- **GPT-4o-mini:**

- **Mixtral (**with fact selection algorithm**):**

Contribution

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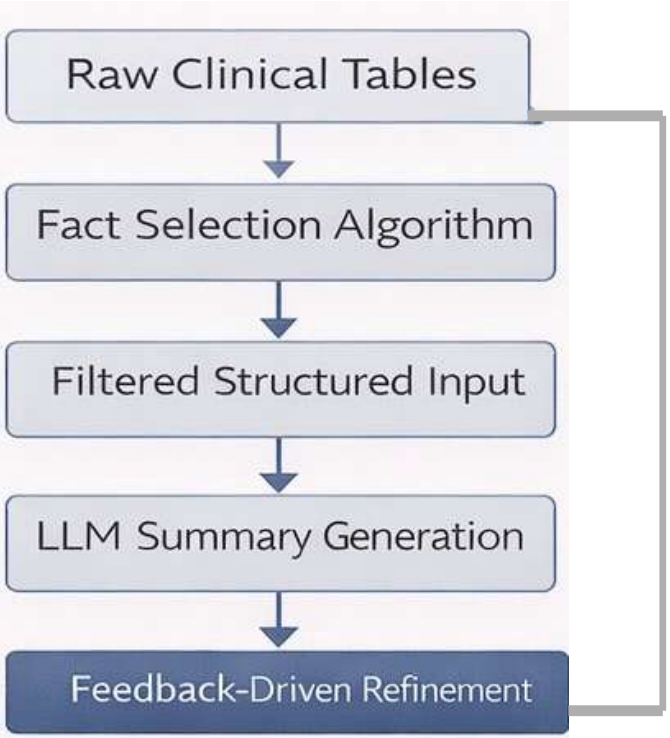


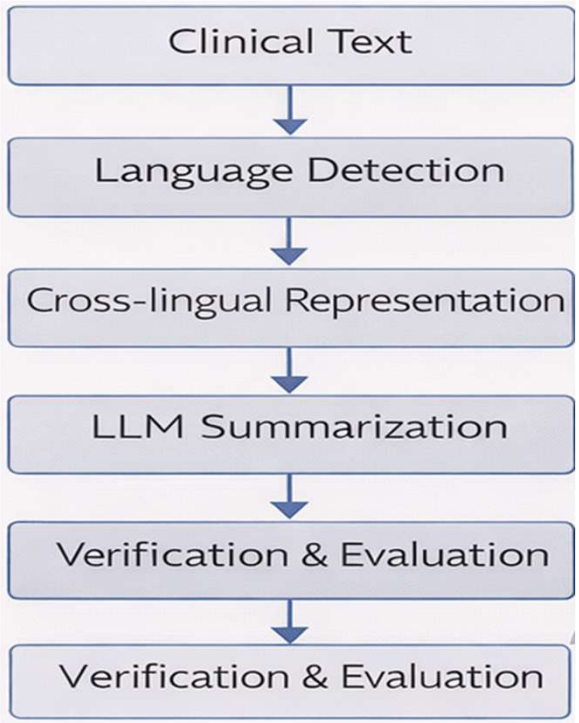
Multilingual Clinical Dialogue Summarization

Problem

- Clinical documentation exists across multiple languages and linguistic contexts
- Cross-lingual summarization errors can lead to unsafe clinical interpretations.
- Translation inconsistencies introduce ambiguity in medical terminology.

Key Idea

- Develop multilingual summarization pipeline preserving clinical meaning across languages
- Evaluate cross-lingual consistency and factual preservation.



Contribution

- Framework for trustworthy multilingual clinical dialogue summarization.
- Improves semantic alignment across languages.

Experimental Results: Multilingual Clinical Dialogue Summarization

Evaluation Setup

- Evaluated on NLP-AI4Health multilingual clinical dialogue dataset.
- Tasks: question answering, summarization, structured information extraction.
- Metrics: Macro F1, ROUGE-L, BERTScore-F1, key-value F1.

Quantitative Results

Language	QnA F1	ROUGE-L	BERT-F1	KNV F1
Marathi	0.23	0.17	0.81	0.30
Kannada	0.47	0.17	0.83	0.27
Gujarati	0.50	0.17	0.84	0.27
English	0.67	0.19	0.84	0.34
Assamese	0.53	0.18	0.83	0.29
Telugu	0.35	0.18	0.83	0.26
Tamil	0.44	0.18	0.84	0.30
Bangla	0.33	0.19	0.82	0.29
Hindi	0.62	0.18	0.84	0.34
Macro Avg.	**0.460**	**0.178**	**0.830**	**0.296**

Findings

- **QnA Performance:** macro F1 of 0.46 shows that model effectively interprets dialogues to answer factual questions.
- **Summarization:** modest ROUGE-L, high BERT-F1 shows strong semantic alignment indicates that generated summaries convey equivalent meaning despite phrasing differences.
- **Structured Extraction:** low F1 score, but still produced some valid schema
- **Language Variability:** languages with high representation in model's pre-training displayed superior performance as compared to low resource languages.

Med-Guard: Safety First Multi-Agent Clinical Decision System

Problem

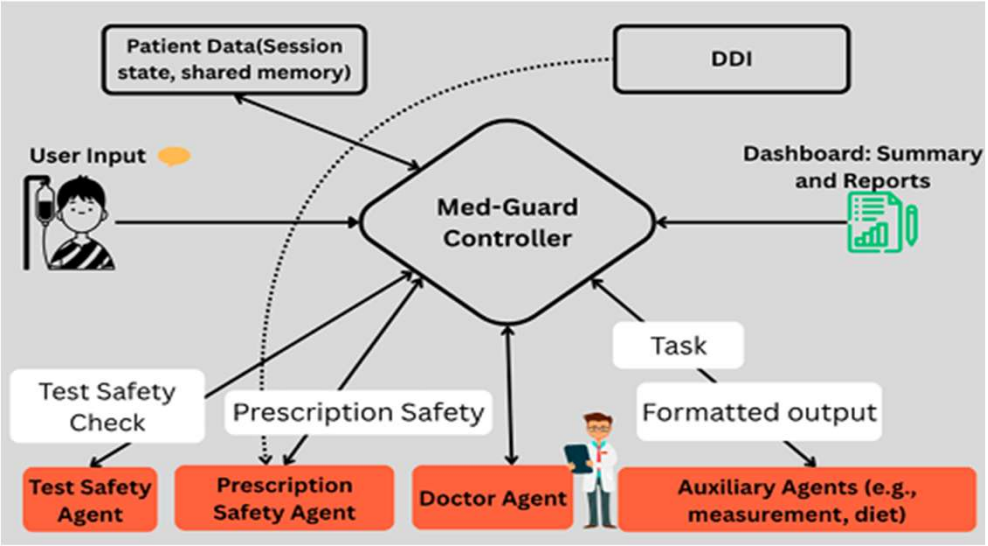
- Diagnostic error contributes to adverse clinical outcomes in healthcare environments.
- Existing LLM based systems face safety risks, hallucinations, and lack of transparency.
- Clinical deployment requires safety verification, auditability, transparency, and structured decision workflows.

Key Idea

- Design a multi-agent LLM framework with explicit safety guardrails.
- Distribute clinical reasoning across specialized agents with defined roles.
- Introduce mandatory safety checks before diagnostic or treatment decisions.
- Introduce differential diagnosis instead of providing single diagnosis.

Contribution

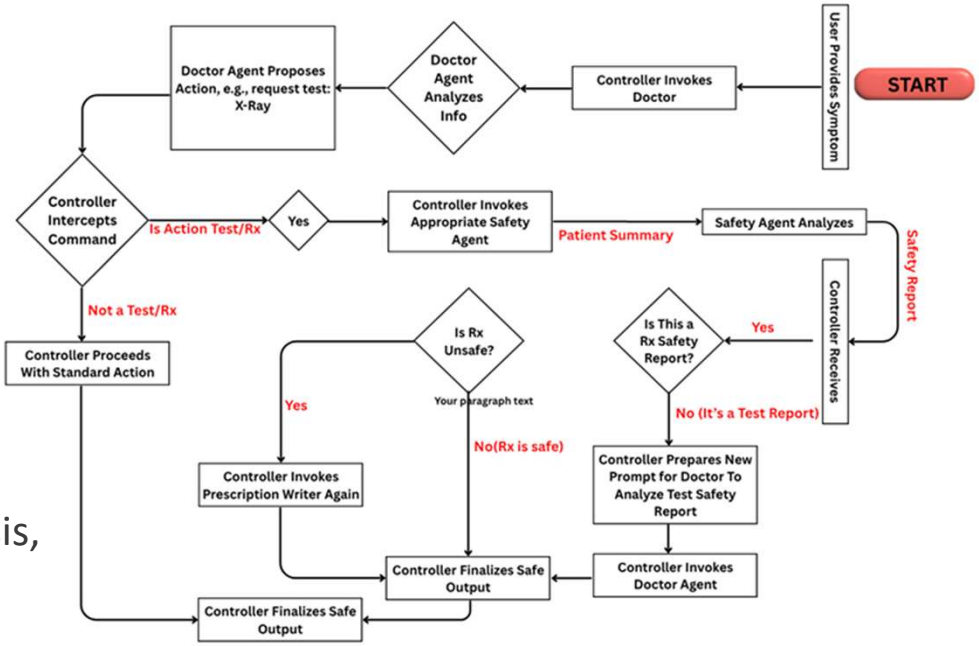
- Safety-first multi-agent architecture for clinical decisions.
- Dedicated safety agents for diagnostic and prescription validation.
- Structured logging and traceability of clinical reasoning steps.



Med-Guard Architecture And Agent Roles

System Design

- Med-Guard uses multiple specialized agents instead of one model.
- A central controller manages workflow and communication.
- Each agent performs a specific clinical task.
- Safety checks are applied before decisions reach the user



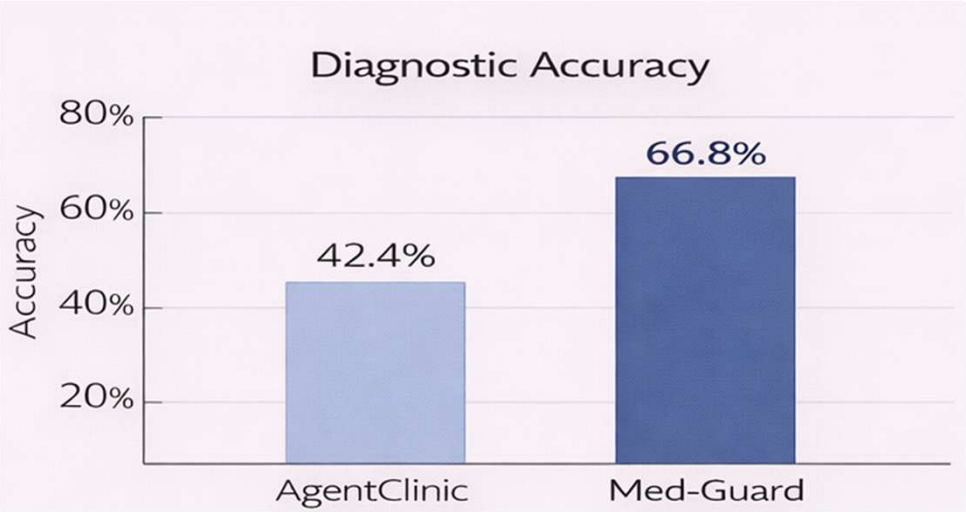
Agent Roles

- **Doctor Agent:** collects symptoms, reasons about diagnosis, suggests tests and treatment.
- **Test Safety Agent:** checks risks of diagnostic tests.
- **Prescription Writer Agent:** drafts medication plan.
- **Prescription Safety Agent:** checks drug interactions, allergies, and contraindications.
- **Measurement Agent:** converts lab results into structured format.
- **Dietary Advisor Agent:** provides lifestyle advice after diagnosis.

Med-Guard Evaluation Results

Evaluation Setup

- Evaluated on AgentClinic-MedQA diagnostic dataset
- Dataset contains clinical cases with symptoms, history, and lab results.
- System tested on diagnostic accuracy and safety.



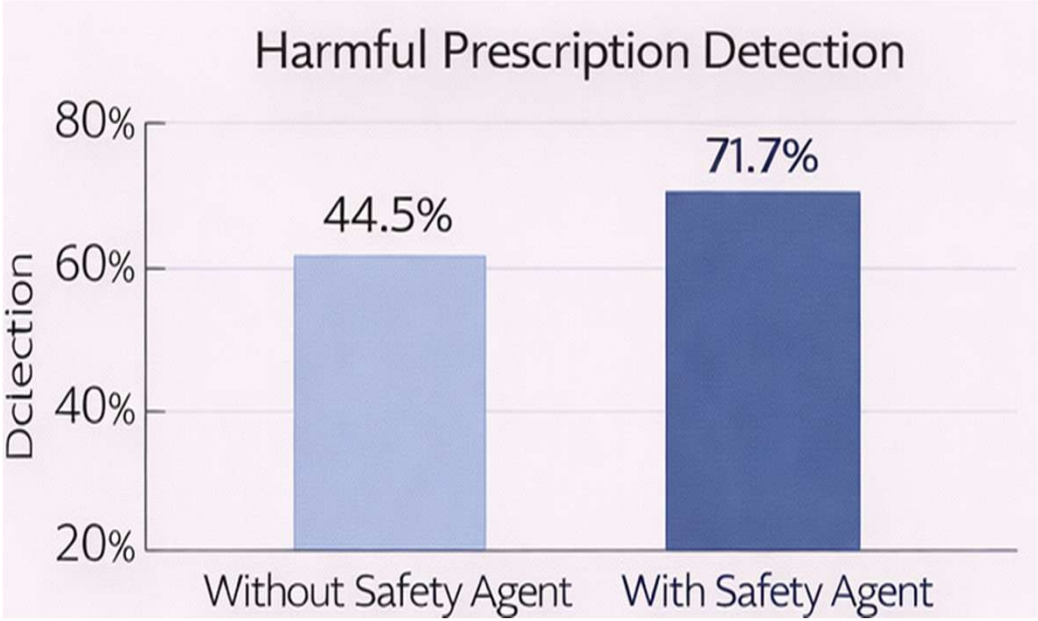
Diagnostic Performance

- **Differential Diagnosis:**
  - Doctor Agent generates a list of candidate diseases.
  - Each candidate is evaluated using patient history and test results.
  - Diagnosis is updated iteratively before final decision.
- Med-Guard **improves diagnostic accuracy** compared to AgentClinic, attributed to differential diagnosis.
- **Example:** Llama-4-Maveric: 66.8% (Med-Guard) vs 42.4% (baseline).
- Multi-turn evaluation reduces premature conclusions.

Med-Guard Safety Agent Design And Impact

Safety Agent Design

- Safety agents verify decisions before they reach the user.
- Independently reviews diagnostic tests and prescriptions.
- Checks for drug-drug interactions.
- Identifies allergy conflicts and contraindications.
- Flags unsafe or high-risk recommendations.
- Prevents unsafe outputs from reaching the patient.



Quantitative Impact

- **Harmful Prescription Detection:**
  - Without safety agent: 44.5%
  - With safety agent: 71.7%

Trust-X: Towards Transparent And Safe Multi-Agent Reasoning

Problem

- Models can be accurate but still reason unreliably.
- Correct answers do not guarantee safe or coherent reasoning.
- Traditional systems lack visibility into intermediate steps.

Contribution

- Introduces Consensus Disagreement Rate (*CDR)* among multiple doctor agents.
- Proposes Epistemic Trust Index (ETI)
- Defines Operational Safety Index (OSI).
- Introduces Final Trust Index (FTI) combining reasoning and safety.

Key Idea

- **Consensus Reasoning:** multiple doctor agents analyze the case independently and vote on the best diagnosis. Their disagreement is recorded as consensus disagreement rate (*CDR*).
- **Reasoning Trace:** each diagnosis step includes a clear trace linking evidence, and reasoning showing how thoughts evolved.
- **End-to-End Logging:** every agent action is recorded with time role and reason, allowing full replay and accountability.
- **Trust Scores:** a trust layer combines interpretability, safety and reasoning quality into measurable trust scores.

Trust-X: Methodology

Multi-Agent Reasoning

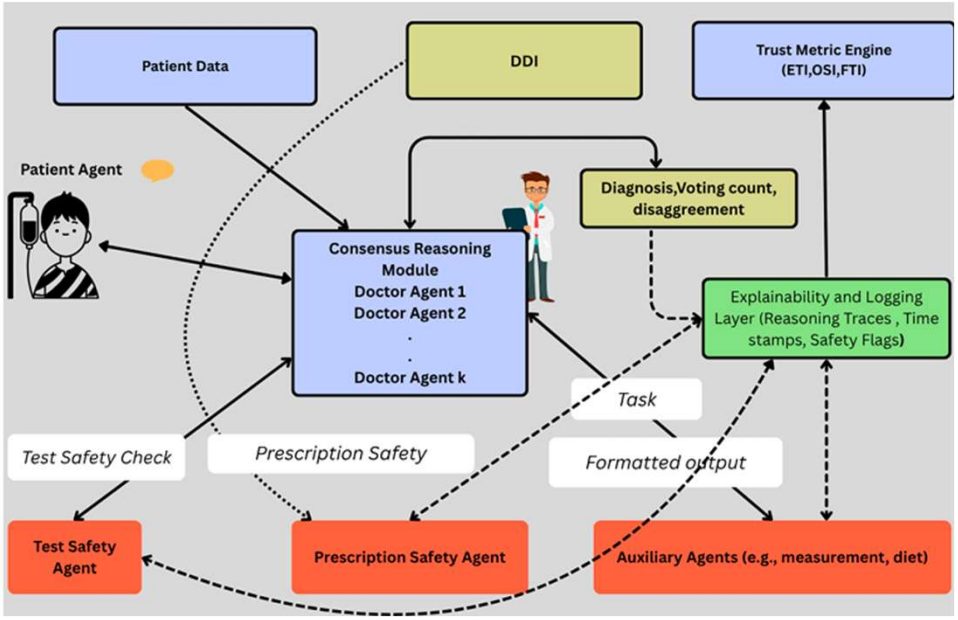
- Three Doctor Agents (K = 3) analyze the same patient case.
- Each produces structured reasoning steps using Differential Diagnosis Lifecycle (DDxL).
- Intermediate hypothesis and discarded diagnoses are logged

Consensus and Uncertainty

- Outputs compared across agents.
- Consensus Disagreement Rate measures variation among diagnoses.
- Higher disagreement indicates higher epistemic uncertainty.
- Alerts and overrides are recorded.

Contribution

- **ETI:** Combines diagnostic accuracy, disagreement, and reasoning coherence.
- **OSI:** penalizes unsafe prescriptions and missed alerts.
- **FTI:** weighted combination of ETI and OSI.



Trust-X: Trust Metrics And Model Comparison

Evaluation Setup

- 50 MedQA clinical cases.
- **Configurations:**
  - **Baseline:** single doctor agent reasoning without consensus or safety checks.
  - **Consensus:** multiple doctor agents reasons. Final decision based on agreement between agents.
  - **Safety:** safety agents are enabled
  - **Trust:** combines consensus reasoning and safety monitoring.

Accuracy

- Accuracy remains similar (~68-69%) across systems, but trust differs sharply when safety supervision is enabled.

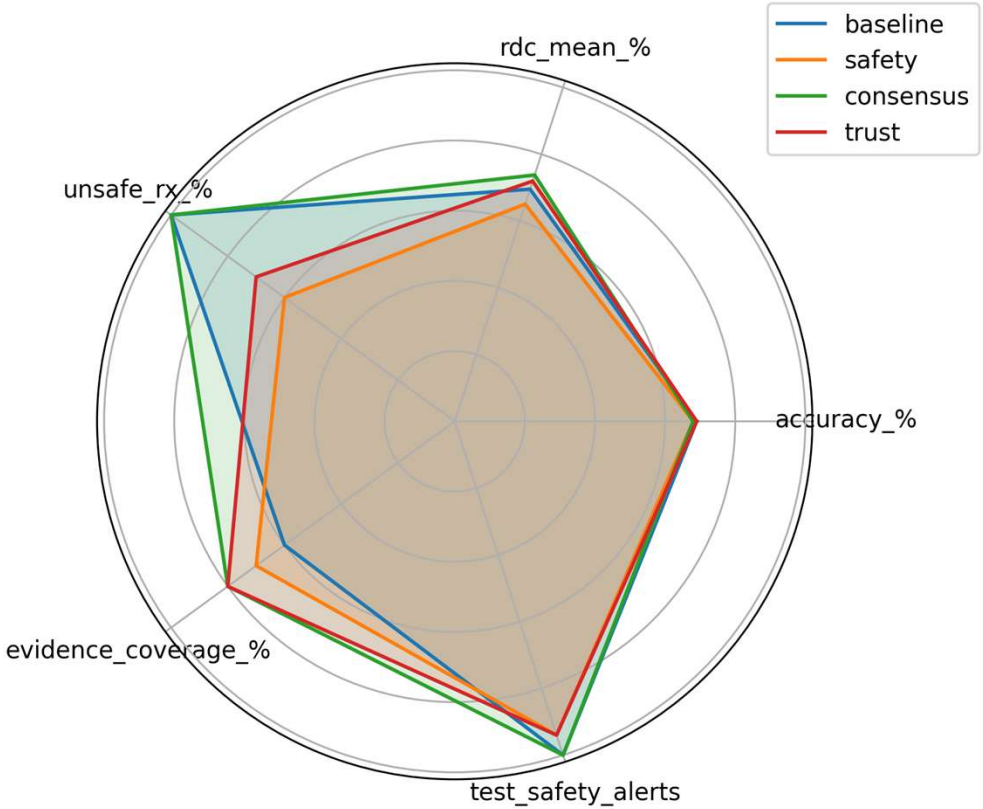
Trust Results

- **Baseline:**
  - OSI = 0 (no safety agents)
  - FTI = 39.2
- **Consensus:**
  - OSI = 0 (no safety agents)
  - FTI = 35
- **Safety:**
  - FTI = 75
- **Trust:**
  - FTI = 75

Trust-X: Multidimensional Trust Profile

Multidimensional Evaluation

- Trust evaluated across multiple behavioral dimensions.
- Metrics include diagnostic accuracy, reasoning coherence, evidence coverage, and safety behavior.
- Safety dimension computed using UnsafeRx and TesAlerts.

Observed Patterns

- **Baseline:** fluent but overconfident reasoning with no explicit safety awareness.
- **Consensus:** improved reflection but limited operational safety.
- **Safety:** cautious reasoning and strong risk detection.
- **Trust:** most balanced profile combining reasoning stability and safety oversight.